$$F[f] = 4\pi F_0 \left(b_0^2 + b_1^2\right) \int_\Omega \left[\hat{g}_k^2 + g_k^2\right] \rho d\rho \quad , F[f] = 4\pi F_0 \left(b_0^2 + b_1^2\right) \int_\Omega \left[\hat{g}_k^2 + g_k^2\right] \rho d\rho$$

$$\frac{1}{\rho}\frac{d}{d\rho}\rho\frac{d}{d\rho}g = -\hat{g} \quad , F[f] = 4\pi F_0 \left(b_0^2 + b_1^2\right) \int_\Omega \left[\hat{g}_k^2 + g_k^2\right] \rho d\rho$$

$$K_0(\rho e^{\pi i/4})] + a_1 \Im[J_0(\rho e^{\pi i/4})] + b_0 \Re[K_0(\rho e^{\pi i/4})] + b_1 \Im[K_0(\rho e^{\pi i/4})] \quad , F[f] = 4\pi F_0 \left(b_0^2 + b_1^2\right) \int_\Omega \left[\hat{g}_k^2 + g_k^2\right] \rho d\rho$$

$$-(1-\gamma_0\rho^2)\left(\rho^2\frac{d^2f}{d\rho^2} - \rho\frac{df}{d\rho}\right) + \rho^4 f = 0 \quad , \lim_{\gamma_0 \to 0} \gamma_0 \left(\frac{df}{d\rho}\right)^2 + \frac{2\rho^3}{\rho}\frac{d^3f}{d\rho^3} + \rho^4\frac{d^4f}{d\rho^4} + \dots$$

$$F(f(\rho)) = 4\pi F_0 \int_{R/\xi}^{\infty} \left[f^2 + \left(\frac{1}{\rho}\frac{d}{d\rho}\rho\frac{d}{d\rho}f\right)^2\right] \rho d\rho \quad , \lim_{\gamma_0 \to 0} \gamma_0 \left(\frac{df}{d\rho}\right)^2 + \frac{2\rho^3}{\rho}\frac{d^3f}{d\rho^3} + \rho^4\frac{d^4f}{d\rho^4} + \dots$$

$$\left(\frac{d^2}{d\rho^2} + \frac{1}{\rho}\frac{d}{d\rho} - i\right)\left(\frac{d^2}{d\rho^2} + \frac{1}{\rho}\frac{d}{d\rho} + i\right)f = 0 \quad , \left(\frac{d^2}{d\rho^2} + \frac{1}{\rho}\frac{d}{d\rho} - i\right)\left(\frac{d^2}{d\rho^2} + \frac{1}{\rho}\frac{d}{d\rho} + i\right)f = 0$$

$$\lim_{\gamma_0 \to 0} \gamma_0 \left(\frac{df}{d\rho}\right)^2 + \frac{2\rho^3}{\rho}\frac{d^3f}{d\rho^3} + \rho^4\frac{d^4f}{d\rho^4} + \dots$$

$$\lim_{\gamma_0 \to 0} \gamma_0 \left(\frac{df}{d\rho}\right)^2 + \frac{2\rho^3}{\rho}\frac{d^3f}{d\rho^3} + \rho^4\frac{d^4f}{d\rho^4} + \dots$$

$$\left(\frac{d^2}{d\rho^2} + \frac{1}{\rho}\frac{d}{d\rho} - i\right)\left(\frac{d^2}{d\rho^2} + \frac{1}{\rho}\frac{d}{d\rho} + i\right)f = 0 \quad , \left(\frac{d^2}{d\rho^2} + \frac{1}{\rho}\frac{d}{d\rho} - i\right)\left(\frac{d^2}{d\rho^2} + \frac{1}{\rho}\frac{d}{d\rho} + i\right)f = 0$$

$$F(f(\rho)) = 4\pi F_0 \int_{R/\xi}^{\infty} \left[f^2 + \left(\frac{1}{\rho}\frac{d}{d\rho}\rho\frac{d}{d\rho}f\right)^2\right] \rho d\rho \quad , F(f(\rho)) = 4\pi F_0 \int_{R/\xi}^{\infty} \left[f^2 + \left(\frac{1}{\rho}\frac{d}{d\rho}\rho\frac{d}{d\rho}f\right)^2\right] \rho d\rho$$

$$-(1-\gamma_0\rho^2)\left(\rho^2\frac{d^2f}{d\rho^2} - \rho\frac{df}{d\rho}\right) + \rho^4 f = 0 \quad , -(1-\gamma_0\rho^2)\left(\rho^2\frac{d^2f}{d\rho^2} - \rho\frac{df}{d\rho}\right) + \rho^4 f = 0$$

$$K_0(\rho e^{\pi i/4})] + a_1 \Im[J_0(\rho e^{\pi i/4})] + b_0 \Re[K_0(\rho e^{\pi i/4})] + b_1 \Im[K_0(\rho e^{\pi i/4})] \quad , K_0(\rho e^{\pi i/4})] + a_1 \Im[J_0(\rho e^{\pi i/4})] + b_0 \Re[K_0(\rho e^{\pi i/4})] + b_1 \Im[K_0(\rho e^{\pi i/4})]$$

$$\frac{1}{\rho}\frac{d}{d\rho}\rho\frac{d}{d\rho}g = -\hat{g} \quad , \frac{1}{\rho}\frac{d}{d\rho}\rho\frac{d}{d\rho}g = -\hat{g}$$

$$\frac{1}{\rho}\frac{d}{d\rho}\rho\frac{d}{d\rho}\hat{g} = g \quad , \frac{1}{\rho}\frac{d}{d\rho}\rho\frac{d}{d\rho}\hat{g} = g$$

$$\frac{1}{\rho}\frac{d}{d\rho}\rho\frac{d}{d\rho}f(\rho) = a_0\Re[J_0(\rho e^{\pi i/4})] \quad , \frac{1}{\rho}\frac{d}{d\rho}\rho\frac{d}{d\rho}f(\rho) = a_0\Re[J_0(\rho e^{\pi i/4})]$$

$$\sqrt{2}F_0/\xi \quad , \nu_L = \sqrt{2F_0/\xi}$$

$$\text{So} \quad , \nu_L = \sqrt{2F_0/\xi}$$

$$\hat{g}_k = \hat{g}_{k-1} - g_{k-1} \quad , n(\rho) = \hat{g}_k' g_k - g_k' \hat{g}_k$$

$$\hat{g}_k = b_0 g_k + b_1 \hat{g}_k \quad , n(\rho) = \hat{g}_k' g_k - g_k' \hat{g}_k$$

$$g_k = a_0 g_b + a_1 \hat{g}_b \quad , n(\rho) = \hat{g}_k' g_k - g_k' \hat{g}_k$$

$$S(b_0^2 + b_1^2)N \to 1/\sqrt{2} \quad , n(\rho) = \hat{g}_k' g_k - g_k' \hat{g}_k$$

$$N = n(\rho_{\text{sat}}) \quad , N = n(\rho_{\text{sat}})$$

$$F[f] = 4\pi F_0 \frac{R}{\xi} \left[S(b_0^2 + b_1^2)N\right] \quad , N = n(\rho_{\text{sat}})$$

$$\lim_{\gamma_0 \to 0} \gamma_0 \left(\frac{df}{d\rho}\right)^2 + \frac{2\rho^3}{\rho}\frac{d^3f}{d\rho^3} + \rho^4\frac{d^4f}{d\rho^4} + \dots \quad , \lim_{\gamma_0 \to 0} \gamma_0 \left(\frac{df}{d\rho}\right)^2 + \frac{2\rho^3}{\rho}\frac{d^3f}{d\rho^3} + \rho^4\frac{d^4f}{d\rho^4} + \dots$$

$$F[f] = 4\pi F_0 \left(b_0^2 + b_1^2\right) \int_\Omega \left[\hat{g}_k^2 + g_k^2\right] \rho d\rho \quad , F[f] = 4\pi F_0 \left(b_0^2 + b_1^2\right) \int_\Omega \left[\hat{g}_k^2 + g_k^2\right] \rho d\rho$$